# CSE200: Online-1 on LaTeX (C1)

## Instructions

- Reproduce the following article using LaTeX exactly as it is.
- You do not need to reproduce this instruction page.
- Ensure that all text formatting, lists, tables, equations, figures,references are implemented as presented in the article with the appropriate LaTeX commands.

## Mark Distribution

| **Component** | **Marks** |
|---|---|
| Text Formatting | 15 |
| Lists | 15 |
| Equations | 25 |
| Tables | 25 |
| Figures | 20 |
| **Total** | **100** |

# CSE200: Online-1 on LaTeX

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# 1 Introduction

Clear technical documentation depends on **precise wording** and *logical structure* to present ideas effectively. This section demonstrates how **written explanations**, *tabular data*, mathematical notation, and visual elements can be combined to produce a well-organized report.

Using **meaningful headings**, *systematic lists*, and important terms improves readability and flow. In addition, controlled use of font size changes, inline symbols ($x_{ij}, y^2$), and consistent spacing contributes to a clean and professional document layout.

# 2 Key Concepts

## 2.1 Advanced Nested and Mixed Lists

1. **Project Setup**
   - @ Install dependencies
   - Configure environment
     (a) Verify Python version
     (b) Check library versions
         * numpy updated

    * pandas updated

2. **Data Processing**

   (a) Load data

   (b) Clean data

      :p: Remove nulls

      :q: Correct formats

         i. Normalize numeric columns

         ii. Encode categorical columns

3. **Result Generation**

   - Save processed data
   - Create reports

     (a) Generate charts

     (b) Export summary CSV

# 3 Mathematical Expressions

$$U = \frac{1}{2}LI^2 + \frac{Q^2}{2C} \tag{1}$$

$$\begin{aligned} h(t) &= e^{-2t}\tan(3t) + \sqrt{t+1} \\ &= \frac{e^{-3t}\sin(5t) + \sqrt{t+2}}{1+t^4} \end{aligned} \tag{2}$$

$$K = \int_0^{\pi} (5\sin x + 4\cos x) dx$$

Piecewise function:

$$f(x) = \begin{cases} x^2 + 3x & \text{if } x \leq 0 \\ \ln(x+2) & \text{if } 0 < x < 2 \\ \frac{1}{x^2+1} & \text{if } x \geq 2 \end{cases}$$

$$S(y) = \sum_{k=0}^{\infty} \frac{(-1)^k y^{2k+1}}{(2k+1)!} + \frac{\ln(y+3)}{y^2+1} - \sqrt{y+5} + \cos(y) \tag{3}$$

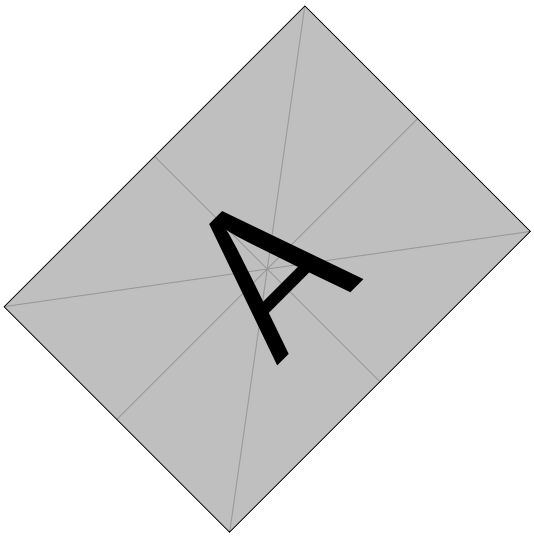
# 4 Comprehensive Performance Analysis

| **Module** | **Stage** | **Sub-module** | **Status** |
|---|---|---|---|
| Alpha | Stage 1 | A1 & Needs Improvement | |
| Alpha | | A2 | Good |
| Beta | Stage 2 | B1 | Good |
| Beta | | B2 | Excellent |
| Gamma | | G1 | Excellent |
| Delta | Stage 3 | D1 | Good |
| Delta | | D2 | Review |
| **Overall Modules** | | **All Sub-modules** | **Satisfactory** |

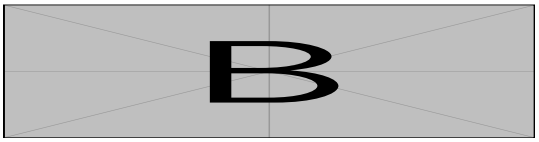
Table 1: Module-wise performance with the Stage column in the middle and multirow usage

Table 1 shows the detailed scores of various modules and their sub-modules across two stages. The final row provides a comprehensive overall performance assessment.

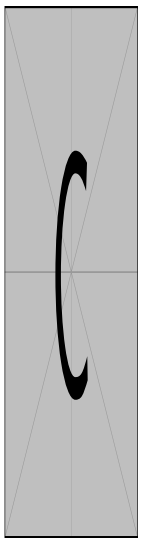
# 5 Diamond-Shaped figures



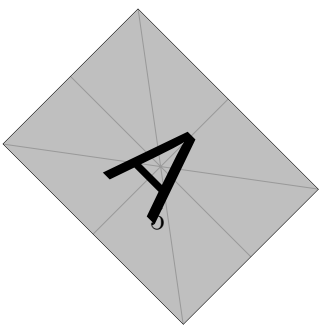

(a) Top




(b) Left




(c) Right




(d) Bottom

Figure 1: Diamond-shaped arrangement

As shown in Figure 1, the images are arranged in a **diamond pattern**: the top and bottom images are centered, while the middle row contains two images aligned slightly left and right. This layout demonstrates **subfigure alignment, scaling, and spacing** in LaTeX.

# Conclusion

In conclusion, effective use of LaTeX allows technical documents to be presented in a clear and structured manner. Combining **well-formatted text**, *tables*, equations, and visual elements improves both readability and organization.

Attention to layout consistency and formatting details plays a key role in producing professional-quality reports. By following these principles, students can ensure that their work is accurate, organized, and easy to understand.